Introduction to Relational Query Languages

RELATIONAL CALCULUS MOTIVATION

Set of Tuples

Set-former notation:

Example

$$\left\{ p.name, p.price \middle| Product(p) \wedge \exists m \left(p \right. \right.$$

Product

name	price	manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper

Manufacturer

name	address	city	state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

RELATIONAL ALGEBRA MOTIVATION

Familiar: Algebra on Numbers

- $(x + 3) - (x \times y)$

- *Variables & values*
- *Operations*
- *Equivalence properties*

Algebra on Relations

Operations such as

- Filtering for tuples meeting condition
- Joining two tables
- Set operations

Equality properties such as commutativity

In Practice

SQL core = RC + RA

SQL query optimization based upon RA and equality properties.